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Research Interests (Keywords)

Cognitive neuroscience of language; aging and language attrition; language–memory interactions; psycholinguistics; Chinese sentence and lexical-semantic processing; statistical modelling of behavioral/EEG data; reproducible research; Chinese NLP; lexical ambiguity and homonymy; human–LLM comparison; interpretable evaluation and benchmarking.

Teaching Interests (SDS)

Statistics and modeling for behavioral/cognitive data (R/Python); experimental design and reproducible research.

Natural language processing for Chinese; LLM evaluation/interpretability; human–AI comparison for language understanding.

Academic Profile

I am a cognitive neuroscientist of language trained in psycholinguistics/neurolinguistics and clinical linguistics. My research investigates Chinese sentence and lexical-semantic processing and how aging-related changes in declarative/procedural memory contribute to language attrition and production–comprehension asymmetries. Methodologically, I combine controlled behavioral experiments (lab and online) with rigorous statistical inference, and where applicable EEG/ERP, to develop mechanistic accounts of language–memory interactions.

In parallel, I build data-centric resources and reproducible computational pipelines for Chinese lexical ambiguity research, using large language models (LLMs) as computational hypotheses and developing interpretable evaluation tasks and open datasets with transparent code and analysis reports. I am prepared to teach effectively in English at undergraduate and graduate levels, including courses in experimental methods, statistics for behavioral/data science, and NLP/LLM evaluation for language data.

Academic Appointments

Postdoctoral Fellow (博士後研究員) — The Hong Kong Polytechnic University (PolyU)

Research Center of Language, Cognition and Neuroscience; Department of Language Science and Technology · 2024–present

- Research design and experimentation; institutional and clinical collaborations
- Conference session organization and chairing
- Grant applications (e.g., HK GRF) and postgraduate exchange coordination

Research Associate (助理研究員) — The Hong Kong Polytechnic University

2023–2024

- Research dissemination and publication
- Proposal development (e.g., NSFC)
- Support for PhD student training and supervision

Research Assistant (研究助理) — The Hong Kong Polytechnic University

2018–2019

- Research on Chinese *bǎ*-construction; data collection and analysis; manuscript preparation
- Contributed to two research papers

Education

Ph.D. in Neuro-cognitive Studies of Language and Clinical Linguistics — The Hong Kong Polytechnic University

2019–2023 · Supervisor: Prof. William Shiyuan WANG

Dissertation: *An experimental investigation into older adults' production/comprehension asymmetries and declarative/procedural memory contributions: a Chinese context*

M.A. in Linguistics and Applied Linguistics — Nankai University

2013–2016

B.A. in Teaching Chinese as a Second Language — Nanjing University of Information Science & Technology

2009–2013

Research Program (3–5 years)

1. **Mechanisms of language attrition in older adults:** Behavioral and (where applicable) neuroimaging studies of aging-related changes in the tip-of-the-tongue phenomenon, with a focus on the retrogenesis/age-of-acquisition effect.
2. **Human–LLM comparison for Chinese lexical ambiguity:** Interpretable tasks and evaluation protocols quantifying how context and grammatical cues drive disambiguation in humans versus language models.
3. **Open resources & reproducible pipelines for cumulative research:** Normative databases (e.g., meaning-specific AoA, meaning dominance) and benchmark datasets for Chinese homonymy/polysemy, released with code and reproducible statistical reports to support student training and community reuse.

Open Research Artifacts (Code & Data)

- **[EMNLP 2025 Main]** *Context and POS in Action: Chinese Homonym Disambiguation in Humans and LLMs* — official implementation and reproducible analysis pipeline.
  https://github.com/neurothew/context-and-pos-in-action
  Contents: LLM judgment generation notebooks; angular similarity computation; stimuli/data; reproducible statistical report (R Markdown).
- **[ACL 2025 Findings]** *Exploring Layer-wise Representations of English and Chinese Homonymy in Pre-trained Language Models* — official implementation (computation + statistics).

https://github.com/neurothew/exploring-homonym-rep-in-llm
Contents: computation scripts; data tables; results; reproducible analysis (R Markdown).

- **Meaning-specific AoA of 562 Chinese homonyms (ongoing normative database)** — dataset repository (MIT license).
  https://github.com/Patrickbigbig/meaning-specific-AoA-of-562-Chinese-homonyms
  Planned release: meaning-specific AoA norms (562 homonyms); anonymized raw ratings (N=160); R Markdown analyses.

Selected Publications (Representative)

1. **Xie, C.*** , Ma, M. K.-H.* , Wang, W., & Wang, W. S. (2025). *Context and POS in Action: A Comparative Study of Chinese Homonym Disambiguation in Human and Language Models.* **Proceedings of EMNLP 2025 (Main Conference)**.
   https://openreview.net/pdf?id=BdIeQMaKQF
2. Ma, M. K.-H.* , **Xie, C.*** (co-first author), Wang, W., & Wang, W. S. (2025). *Exploring Layer-wise Representations of English and Chinese Homonymy in Pre-trained Language Models.* **Findings of ACL 2025**, 19705–19724.
   https://aclanthology.org/2025.findings-acl.1011/
3. **Xie, C.**, & Wang, W. S. (2026). *The Tip-of-the-Tongue Phenomenon: Cognitive, Neural, and Neurochemical Perspectives.* **Biomedicines**, 14(2), 269.
   https://doi.org/10.3390/biomedicines14020269
4. Ma, M. K.-H., Fong, M. C.-M., **Xie, C.**, Lee, T., Chen, G., & Wang, W. S. (2021). *Regularity and randomness in ageing: Differences in resting-state EEG complexity measured by largest Lyapunov exponent.* **NeuroImage: Reports**, 1(4), 100054.
   https://doi.org/10.1016/j.ynirp.2021.100054
5. **Xie, C.**, Fong, M. C.-M., Ma, M. K.-H., Wang, J., & Wang, W. S. (2023). *The retrogenesis of age-related decline in declarative and procedural memory.* **Frontiers in Psychology**, 14, 1212614.
   https://doi.org/10.3389/fpsyg.2023.1212614

* equal contribution

Peer-Reviewed Conference Papers

- **Xie, C.*** , Ma, M. K.-H.* , Wang, W., & Wang, W. S. (2025). *Context and POS in Action: A Comparative Study of Chinese Homonym Disambiguation in Human and Language Models.* **EMNLP 2025 (Main Conference)**, Suzhou, China.
  https://openreview.net/pdf?id=BdIeQMaKQF
- Ma, M. K.-H.* , **Xie, C.*** (co-first author), Wang, W., & Wang, W. S. (2025). *Exploring Layer-wise Representations of English and Chinese Homonymy in Pre-trained Language Models.* **Findings of ACL 2025**, Vienna, Austria.
  https://aclanthology.org/2025.findings-acl.1011/
- **Xie, C.**, Feng, Y., & Wang, W. S. (2020). *Association between declarative memory and language ability in older Chinese by education level.* **Proceedings of the 34th Pacific Asia Conference on Language, Information and Computation (PACLIC 34)**, Hanoi, Vietnam.

Peer-Reviewed Journal Articles

- **Xie, C.**, & Wang, W. S. (2026). *The Tip-of-the-Tongue Phenomenon: Cognitive, Neural, and Neurochemical Perspectives.* **Biomedicines**, 14(2), 269. https://doi.org/10.3390/biomedicines14020269
- 謝郴偉 (**Xie, C.**). (2025). *"王理论"：语言马赛克观与神经语言学的多元整合* (William Wang Principle: Mosaic Theory and Its Multidimensional Integration in Neurolinguistics). **實驗語言學 (Experimental Linguistics)**, 14(3).

- 謝郴偉 (**Xie, C.**), & Wang, W. S. (2025). *语言学中的演化视角* (An evolutionary view in linguistics). **實驗語言學 (Experimental Linguistics)**, 14(1), 55–62.
- **Xie, C.**, Fong, M. C.-M., Ma, M. K.-H., Wang, J., & Wang, W. S. (2023). *The retrogenesis of age-related decline in declarative and procedural memory.* **Frontiers in Psychology**, 14, 1212614. https://doi.org/10.3389/fpsyg.2023.1212614
- Ma, M. K.-H., Fong, M. C.-M., **Xie, C.**, Lee, T., Chen, G., & Wang, W. S. (2021). *Regularity and randomness in ageing: Differences in resting-state EEG complexity measured by largest Lyapunov exponent.* **NeuroImage: Reports**, 1(4), 100054. https://doi.org/10.1016/j.ynirp.2021.100054
- **Xie, C.**, Feng, Y., et al. (2020). *Association between declarative memory and language ability in older Chinese by education level.* **實驗語言學 (Experimental Linguistics)**.
- 謝郴偉 (**Xie, C.**). (2019). *广州话中低平调混淆的感知研究* (Perception research on the confusion of mid-level and low-level tones in Cantonese). **中国语音学报 (Chinese Journal of Phonetics)** (1).
- 謝郴偉 (**Xie, C.**). (2019). *"将"字句韵律分布模式研究* (Prosodic distribution pattern of the *jiāng*-construction). **汉语学习 (Chinese Language Learning)**.
- 温宝莹 (Wen, B. Y.), & 謝郴偉 (**Xie, C.**) (2018). *日本学习者汉语陈述句语调的韵律匹配* (Prosodic matching of intonation in Chinese declaratives by Japanese learners). **南开学报：哲学社会科学版** 0(4).
- 謝郴偉 (**Xie, C.**). (2018). *双音节动词"把"字句的焦点考察* (Focus in *bǎ*-construction with disyllabic verbs). **International Journal of Chinese Linguistics**.
- 丛珊 (Cong, S.), & 謝郴偉 (**Xie, C.**) (2017). *鄂伦春语的元音格局及其高元音的声学分析* (Vowel system of Oroqen and acoustic analysis of high vowels). **中央民族大学学报：哲学社会科学版** 0(6).
- 謝郴偉 (**Xie, C.**). (2017). *"被"字句和"把"字句中处置义的差异——韵律分布模式的证据* (Differences in disposal meaning between *bèi*/*bǎ* constructions: evidence from prosody). **南开语言学刊 (Nankai Linguistics)**.
- 謝郴偉 (**Xie, C.**), 石锋 (Shi, F.), et al. (2017). *广州话平调的感知* (Perception of level tones in Cantonese). **清华大学学报：自然科学版 (Journal of Tsinghua University: Natural Science Edition)**, 57(3).
- 謝郴偉 (**Xie, C.**), et al. (2015). *广州话双字组平调的听感实验初探* (Perception of level tones in Cantonese disyllables). **中国语音学报 (Chinese Journal of Phonetics)**, 5(1).
- 謝郴偉 (**Xie, C.**). (2015). *基于语义的HSK甲级汉日同形形容词难度等级研究* (Difficulty levels of identical-form Chinese–Japanese adjectives in HSK Level A). **现代语文：语言研究** (4), 106–108.
- 謝郴偉 (**Xie, C.**), & 徐开妍 (Xu, K.) (2013). *基于HSK甲级形容词的汉日同形词研究与对日汉语词汇教学* (Chinese–Japanese identical-form words and implications for teaching). **文教资料** (10), 68–69.

Book Chapters

- **Xie, C.** (2023). *Aging Effects on Semantic and Syntactic Abilities in the Chinese Population.* In J. Kong, G. Peng, Z. Shen, & W. Feng (Eds.), *Inspiration from a Lofty Mountain: Festschrift in Honor of Professor William S.-Y. Wang on his 90th Birthday* (pp. 370–382). City University of Hong Kong Press.
- 謝郴偉 (**Xie, C.**). (2021). *双音节动词"把"字句的焦点考察* (Focus in *bǎ*-construction with disyllabic verbs). In 石锋 (Ed.), *韵律格局：语音和语义、语法、语用的结合* (pp. 220–235). Beijing: The Commercial Press.

Manuscripts in Preparation

- **Xie, C.**, Feng, Y., Ma, M. K.-H., Wang, W., & Wang, W. S. (in preparation). *Retrogenesis in Meaning Selection: Age, Context Similarity, and POS Effects on Homonym Resolution.* Target journal: *Aging, Neuropsychology, and Cognition*.
- **Xie, C.**, Ma, M. K.-H., Wang, W., & Wang, W. S. (in preparation). *Meaning dominance and meaning-specific age of acquisition for Chinese disyllabic words: A normative database for homonyms, polysemes, and monosemes.* Target journal: *Behavior Research Methods*.

Grants & Funded Projects

Funded

- **On the tip of the tongue: the impact of AoA on word retrieval difficulty in normal ageing**
  Role: **Co-Investigator** · Funder: Departmental Earnings (internal), Department of Language Science and Technology, PolyU · Amount: **HKD 400,000** · Period: **2024–2026**
- **Longitudinal tracking of lifestyle and cognitive changes based on actigraphy and electroencephalography**
  Role: **Co-Investigator** · Funder: Research Institute for Smart Ageing (RISA) Projects (internal) · Amount: **HKD 300,000** · Period: **2024–2027**

Submitted (Under Review)

- **Language Learning across the Lifespan: An fMRI–fMRS Study of Functional and Neurochemical Measures in Cerebellar Plasticity**
  Submitted to: **Hong Kong RGC GRF 2026/2027** · Submission year: **2025** · Role: **Co-Investigator** · Outcome: **Under review**
- **Aligning Language Models with the Brain: Hierarchical Mechanisms of Lexical and Structural Ambiguity Resolution**
  Submitted to: **RGC Junior Research Fellow Scheme 2026/27** · Submission year: **2025** · Role: **Candidate / PI-equivalent** · Outcome: **Under review**

Invited Talks & Public Lectures

- **Invited Speaker**, "Experimental Linguistics+" Live Forum (2025). Public lecture: *Humans, Models, and Meaning* (4,000+ online views).

Conference Presentations

- **Xie, C.**, Ma, M. K.-H., Wang, W., & Wang, W. S. (2025). *Probing Neurodegeneration and Aging Through iPSC-Derived Models.* The 16th International Conference in Evolution Linguistics, Yanbian, China.
- **Xie, C.** (2024). *Exploring the Multifaceted Challenges of Aging: Unveiling Genetic, Neural, and Alzheimer's Insights in China and Beyond.* The 15th International Conference in Evolution Linguistics, Changsha, China.
- **Xie, C.** (2023). *Why is conversation not easy for older adults? An experimental study on production/comprehension asymmetry in Chinese older adults.* The 4th Macau Symposium on Cognitive and Brain Sciences, Macau SAR, China. **(Best Oral Presentation Award)**
- **Xie, C.** (2023). *Aging effects on the language abilities of the oldest old.* The 14th International Conference in Evolution Linguistics, Hong Kong SAR, China.
- **Xie, C.**, & Wang, W. S. (2023). *A return to childhood? The retrogenic production/comprehension asymmetry in Chinese older adults.* The 29th Annual Conference of the International Association of Chinese Linguistics, Macau SAR, China.
- **Xie, C.**, & Wang, W. S. (2023). *Speak more or understand more: A preliminary study on age-related changes in the association between language and memory abilities.* The PolyU Academy for Interdisciplinary Research 2023, Hong Kong SAR, China.
- 謝郴偉 (**Xie, C.**) (2023). *返老还童？基于中国老年人的产出/理解不对称性及其陈述性/程序性记忆机制的实验研究* (Rejuvenation? An experimental study of production/comprehension asymmetry and declarative/procedural memory mechanisms in older Chinese). Young Scholars Forum on the Interdisciplinary Research of Language Intelligence and Linguistics, Changsha, Hunan, China.
- 謝郴偉 (**Xie, C.**), & 王士元 (Wang, W. S.) (2022). *Language Variation across the Lifespan, Forestalling cognitive decline.* Boya Linguistic Forum: Language Variation and Change, Peking University, Beijing, China.
- **Xie, C.**, Feng, Y., & Wang, W. S. (2022). *Production-comprehension asymmetries of syntactic processing in Chinese older adults.* The 28th Annual Conference of the International Association of Chinese Linguistics, Hong Kong SAR, China.
- **Xie, C.**, Fong, C. M., & Wang, W. S. (2022). *Production-comprehension asymmetries of semantic processing in Chinese older adults.* The 29th Annual Meeting of the Cognitive Neuroscience Society, San Francisco, USA.
- **Xie, C.**, Feng, Y., et al. (2021). *Speech recoding in reading sinograms: Evidence from an ERP study on Cantonese.* The 32nd International Congress of Psychology, Prague, Czech Republic.

- 謝郴偉 (**Xie, C.**) (2019). *Age-related differences in speech recoding.* The 11th International Conference in Evolution Linguistics, Shanghai, China.
- 謝郴偉 (**Xie, C.**) (2018). *Sentence processing in Chinese younger and older adults.* The 10th International Conference in Evolution Linguistics, Nanjing, China.
- 謝郴偉 (**Xie, C.**) (2017). *"把"字句的焦点在哪里——语调量化分析的证据* (Where is focus in the *bǎ*-construction? Evidence from quantitative intonation analysis). The 9th International Conference in Evolution Linguistics, Kunming, China.
- 謝郴偉 (**Xie, C.**) (2015). *日本留学生汉语语调习得实验研究* (An experimental study on Japanese learners' acquisition of Chinese intonation). The 1st International Symposium on Prosody, Tianjin, China.
- 謝郴偉 (**Xie, C.**) (2015). *广州话单字音平调的听感范畴* (Perceptual categories of level tones in monosyllabic Cantonese pronunciations). The 2015 Annual Academic Conference of the Tianjin Linguistic Society, Tianjin, China. **(Student Group Third Prize)**
- 謝郴偉 (**Xie, C.**) (2013). *基于语义的HSK甲级汉日同形形容词的难度等级研究与对日汉语词汇教学* (Difficulty levels for identical-form Chinese–Japanese adjectives in HSK Level A based on semantics, and implications for teaching Chinese vocabulary to Japanese learners). The 2013 Annual Academic Conference of the Tianjin Linguistic Society, Tianjin, China. **(Student Group Second Prize)**

Academic Service

- **Reviewer:** *Frontiers in Psychology* (2025)
- **Session Chair:** "Aging and Language Degeneration" Session, 15th International Conference in Evolutionary Linguistics (2024)

Teaching Experience

Invited Teaching

- **Invited Lecturer** (2026). *Evolutionary Linguistics: Phylogeny and Ontogeny*, National Summer Institute in Linguistics (Organizer: Nankai University). Language: Mandarin.

Teaching Assistant (English-medium)

- **Teaching Assistant** (2022). *Advanced Topics in Psycholinguistics and Neurolinguistics* (Postgraduate), PolyU. Duties: tutorials, grading, office hours.
- **Teaching Assistant** (2021). *Contrastive Analysis of Chinese and English* (Undergraduate), PolyU. Duties: tutorials/discussion sections, grading, answer keys and supplementary notes.

Instructor of Record

- **Advanced Spoken Chinese Instructor** (2016). University of Wisconsin–Madison Tianjin Summer Chinese Program. 10 weeks; 4 contact hours/week; class size 5; students: U.S. undergraduates. Duties: course design/delivery, assessment, student support, curriculum coordination, co-curricular activities.
- **Elementary Chinese Instructor** (2015). Nankai University × Aichi University Program. 16 weeks; 8 contact hours/week; class size 12; students: Japanese university students.

Mentoring

- **Mentor**, PolyU Junior Research Mentoring Programme (2021)

Awards & Honors

- **Best Oral Presentation Award**, The 4th Macau Symposium on Cognitive and Brain Sciences (2023)
- **Top Ten Graduate ("南開十傑")**, Nankai University (2016)

Methods & Technical Skills

- **Programming / Data:** R, Python, JavaScript; reproducible reporting (R Markdown)
- **NLP / LLM Tooling:** Hugging Face Transformers; HanLP; corpus/text analytics
- **Data Visualization:** ggplot2, matplotlib, Gephi
- **Experiment & Speech Tools:** PsychoPy, lab.js, Praat
- **Experimental Methods:** behavioral experiments; EEG/ERP; online data collection
- **Languages:** Mandarin (native), English (fluent), Japanese (intermediate)